

Mihir Mankikar

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EDUCATION

Texas A&M University , College Station, TX	Aug. 2024 – May 2028
Bachelor of Science in Computer Science, Minor in Statistics	GPA: 3.9
Engineering Honors; EH: Enhancing the Development of Global Engineers	

EXPERIENCE

Texas Instruments	Jun. 2026 – Aug. 2026
<i>AI/ML Automation Intern</i>	<i>Dallas, TX</i>

- Overhauled an AI-powered test-floor log analyzer, replacing vector embeddings with keyword extraction and a GraphRAG knowledge graph storing reference documents and historical event nodes linked by correlator edges to save engineer-hours through automated root-cause analysis
- Engineered an end-to-end agentic AI pipeline with custom Python Model Context Protocol (MCP) tools for Claude to explore enterprise SQL schemas, generate context-driven documentation, and autonomously plan, generate, review, and iterate on code
- Built a compliance agent that analyzes 55+ fields from 10 sources across 12 classification levels using a YAML-driven rules engine deployed via Docker and nginx to automatically validate soon-to-be released devices against internal datasheets
- Architected a Kafka-triggered n8n workflow pipeline for automated semiconductor process document review, cross-referencing content against internal systems to surface incorrect specification references and content errors before human review and prevent scrap incidents
- Developed a React SPA cataloging team tools with dynamic JSON-driven rendering, hosted on an internal Linux server, with an Express proxy containerized separately on Docker that injects internal context and forwards chat requests to an on-prem model

Texas A&M Rocket Engine Design	Jan. 2026 – Present
<i>Avionics and Controls Systems Member</i>	<i>College Station, TX</i>

- Designed robust embedded communications infrastructure for “Elysium 2” liquid bipropellant rocket engine with automatic retry logic, per-packet state tracking, deterministic failure analysis, and state-dependent auto-abort
- Reduced valve command overhead by 74% and overall protocol payload by up to 80% by migrating flight control communications from serial to binary protocol across graphical interface (Python) and Teensy 4.1 firmware (C++)
- Engineered dual-mode fail-safe system in C++ and Python with heartbeat-driven connection health monitoring and measured recovery diagnostics to distinguish transient from persistent communication loss

Texas A&M Sketch Recognition Lab	Feb. 2026 – May 2026
<i>Machine Learning Researcher</i>	<i>College Station, TX</i>

- Collaborated with the U.S. Space Force to build a custom PyTorch hybrid CNN classifying u-blox receiver data as Clean, Spoofing, or Jamming, fusing dual-channel RF spectrum with 30 navigation features
- Achieved 0.90 spoofing and 1.00 jamming recall on a real controlled-attack dataset of 1.6M one-second samples
- Designed a dual-branch architecture (residual CNN with Squeeze-and-Excitation attention plus an MLP) trained with Focal Loss and weighted oversampling to overcome a severe 19:1 class imbalance
- Built the full monitoring stack with a Flask backend and interactive frontend streaming live model outputs, system health, satellite metrics, and confidence scores, plus gradient-based feature attribution for per-prediction interpretability

Aggie Research Program	May 2025 – Present
<i>Machine Learning Researcher</i>	<i>College Station, TX</i>

- Developed a univariate time-series forecasting framework (ARIMA, SARIMA) for U.S. cow-calf price prediction, implementing CPI-deflation to real prices and walk-forward validation with expanding windows
- Extended baseline statistical models to SARIMAX by incorporating exogenous economic predictors, utilizing AIC/BIC criteria and grid search for rigorous model selection
- Advanced the predictive framework by engineering complex features (lag structures, Boruta feature selection) to implement multivariate Random Forest, LightGBM, and SVR pipelines with Optuna hyperparameter optimization
- Developed a high-performance LSTM deep learning architecture to capture non-linear temporal dependencies, demonstrating superior accuracy over traditional models for long-term economic sustainability
- Co-authored 2 papers currently under review at major journals; poster at the 2025 Texas A&M Undergraduate Research Exposition

PROJECTS

- Southwest Airlines Weather Impact Score** | *Python, scikit-learn, PyTorch, Streamlit*
Collaboration with Southwest Airlines to predict weather-induced operational disruptions
- Celebrity-Style Insult Generator** | *Python, Flask, aiotextgen, NLTK, Beautiful Soup*
NLP insult generator in the style of a celebrity

SKILLS

- Languages:** Python, C++, SQL, Java, R
- AI/ML:** TensorFlow, PyTorch, scikit-learn, statsmodels, pandas, NumPy
- Tools:** Claude Code, Codex, Git, Docker, LaTeX, Linux/Unix, Shell scripting (Bash/Zsh)
- Interests:** Marathi (Native), Japanese (N5), Piano, Clarinet, Bouldering